TotalSegmentator Local Setup and ProstateX-0204 Run Memo

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Contents

Status: local run memo. This records how TotalSegmentator was installed and run on Martha for the ProstateX-0204 T2 fixture. It complements fa-segment.md, which owns the broader segmentation strategy.

Summary

We installed a local segmentation environment in `.venv-seg`, downloaded one targeted ProstateX-0204 transverse T2 series from TCIA, converted it from DICOM to NIfTI, and ran TotalSegmentator `total_mr` on CPU and GPU.

The best current output for visual QA is:

```
base:    subjects/prostatex-0204/t2.nii.gz
overlay: seg_out/t2-gpu-full/prostate.nii.gz
```

The full GPU run completed successfully on Martha's Quadro T2000, but it used 3621 MiB of 4096

MiB VRAM. That is enough for this small fixture, but close enough to the ceiling that larger volumes may fail on Martha. Rudolf remains the safer target for routine GPU segmentation.

The main result is narrower than “pelvic segmentation works.” TotalSegmentator gives a plausible whole-gland prostate mask plus useful pelvic landmarks. It does **not** provide the two masks required by the Michallek-faithful FA calculation: the internal-obturator muscle calibration ROI and the tumour-interface analysis ROI.

Installation

The environment was created in the repository root:

```
cd /home/lukass/workspace/frac
python3 -m venv .venv-seg
.venv-seg/bin/python -m pip install --upgrade pip
.venv-seg/bin/python -m pip install -r requirements-segmentation.txt
.venv-seg/bin/python -m pip install dcm2niix
```

The pinned segmentation requirements are:

```
torch==2.5.1+cu121
TotalSegmentator==2.13.0
```

The installed converter is:

```
dcm2niix v1.0.20260416
```

`dcm2niix` is available inside the virtual environment as:

```
.venv-seg/bin/dcm2niix
```

It is not assumed to be on the global shell `PATH`.

Local Data Policy

The following local paths are intentionally ignored by Git:

```
.venv-seg/
subjects/
seg_out/
```

`subjects/` contains the downloaded DICOM cache and derived NIfTI input. `seg_out/` contains generated segmentation masks, run manifests, and GPU memory logs.

Data Download

The fixture is patient `ProstateX-0204` from the public TCIA `PROSTATEx` collection. The series list was queried with `tcia_utils`:

```
python3 -c "from tcia_utils import nbia; import json; s=nbia.getSeries(collection='PR
```

Two t2_tse_tra candidates were present. We used the earlier series:

```
PatientID:         ProstateX-0204
Collection:        PROSTATEx
SeriesDescription: t2_tse_tra
SeriesNumber:      5
ImageCount:        21
SeriesInstanceUID: 1.3.6.1.4.1.14519.5.2.1.7311.5101.327326038201137210751244609393
```

Download command:

```
python3 -c "from tcia_utils import nbia; nbia.downloadSeries(['1.3.6.1.4.1.14519.5.2.
```

The downloaded DICOM cache is retained at:

```
subjects/prostatex-0204/dicom/
```

DICOM to NIfTI Conversion

The DICOM series was converted with the virtualenv dcm2niix:

```
.venv-seg/bin/dcm2niix -z y -o subjects/prostatex-0204 -f t2 subjects/prostatex-0204/
```

The resulting input volume is:

```
subjects/prostatex-0204/t2.nii.gz
subjects/prostatex-0204/t2.json
```

NIfTI sanity check:

```
shape: 384 x 384 x 21
zooms: 0.5 x 0.5 x 3.0 mm
```

This is a 3D anatomical T2 volume. It is valid input for TotalSegmentator `total_mr`. The 4D DCE perfusion fixture is not valid input for this run.

Helper Script

Runs used:

```
scripts/try_totalsegmentator.sh
```

The helper:

- validates that the input NIfTI is 3D
- creates/sources `.venv-seg` if needed
- installs `requirements-segmentation.txt` if TotalSegmentator is missing
- runs TotalSegmentator with `--task total_mr`
- writes one mask NIfTI per label

- writes `run.json` provenance
- records GPU memory to `gpu-memory.csv` for GPU runs

The run manifest includes:

```
tool
tool_version
torch_version
python_version
task
mode
scope
fast
input
output_dir
device
command
runtime_s
exit_status
gpu_memory_log
peak_vram_mib
visual_qa_note
```

`mode` records the model speed variant: `fast` or `full`. `scope` records the requested output class scope: `full` or `prostate` when `--prostate-only` is used.

Runs Performed

Four runs were performed and retained.

Output directory	Device	Mode	Runtime	Peak VRAM	Status
`seg_out/t2/`	CPU	fast	13 s	n/a	success
`seg_out/t2-full/`	CPU	full	29 s	n/a	success
`seg_out/t2-gpu-fast/`	GPU	fast	33 s	2547 MiB	success
`seg_out/t2-gpu-full/`	GPU	full	62 s	3621 MiB	success

These are warm-cache runtimes after the TotalSegmentator weights were already present in `~/.totalsegmentator`. The first cold run also downloads model weights and takes longer; the local weight cache is currently about 614 MB for the models used here.

Commands:

```
./scripts/try_totalsegmentator.sh --fast --cpu subjects/prostatex-0204/t2.nii.gz seg_
./scripts/try_totalsegmentator.sh --cpu subjects/prostatex-0204/t2.nii.gz seg_out/t2-
./scripts/try_totalsegmentator.sh --fast subjects/prostatex-0204/t2.nii.gz seg_out/t2
./scripts/try_totalsegmentator.sh subjects/prostatex-0204/t2.nii.gz seg_out/t2-gpu-fu
```

The recommended output for review is the full GPU result:

```
seg_out/t2-gpu-full/prostate.nii.gz
```

CPU and GPU Agreement

The full CPU and full GPU prostate masks are effectively identical for this fixture:

```
CPU full prostate voxels: 50373
GPU full prostate voxels: 50364
Dice coefficient:         0.9999106584472438
Symmetric difference:     9 voxels
```

This small difference is consistent with numerical differences between CPU and GPU inference kernels and is not an anatomical concern.

What Frac Actually Needs

The current FA validation and strategy documents make the mask requirements explicit:

Mask	Required for	Current source	Automation difficulty
Internal-obturator muscle sample	Step 1 intensity standardization	Reviewed DCE-native sample, seeded/landmark proposal on the selected FA reference, or future task-specific helper	Easier; needs a stable interior sample, not a perfect anatomical contour
Tumour-interface ROI	Step 4 mean-FD aggregation and max-mean-FD	Reviewed/imported ROI until the author rule is pinned down	Hard; blocked on lesion contour plus interface-band rule

TotalSegmentator does not emit either of those target masks. This confirms the expectation in fa-segment.md: broad MR segmenters may expose nearby pelvic context labels, but they should not be advertised as internal-obturator muscle ROI providers unless their exact label list and validation prove that target.

What TotalSegmentator contributes now is context:

- whole-gland prostate mask for registration QA, gland context, lesion-search bounds, and later capsular-contact checks
- bladder, femur/hip, iliopsoas, gluteal, and iliac-vessel landmarks that may help a future atlas or landmark-based internal-obturator ROI proposer

Segmentations Produced

The `total_mr` task writes one NIfTI mask per supported label. For this prostate T2 crop, the full GPU run produced 52 files in `seg_out/t2-gpu-full/`: 50 label masks plus `run.json` and `gpu-memory.csv`.

The mask files are:

```
adrenal_gland_left.nii.gz
adrenal_gland_right.nii.gz
aorta.nii.gz
autochthon_left.nii.gz
autochthon_right.nii.gz
brain.nii.gz
clavicula_left.nii.gz
clavicula_right.nii.gz
colon.nii.gz
duodenum.nii.gz
esophagus.nii.gz
femur_left.nii.gz
femur_right.nii.gz
gallbladder.nii.gz
gluteus_maximus_left.nii.gz
gluteus_maximus_right.nii.gz
gluteus_medius_left.nii.gz
gluteus_medius_right.nii.gz
gluteus_minimus_left.nii.gz
gluteus_minimus_right.nii.gz
heart.nii.gz
hip_left.nii.gz
hip_right.nii.gz
humerus_left.nii.gz
humerus_right.nii.gz
iliac_artery_left.nii.gz
iliac_artery_right.nii.gz
iliac_vena_left.nii.gz
iliac_vena_right.nii.gz
iliopsoas_left.nii.gz
iliopsoas_right.nii.gz
inferior_vena_cava.nii.gz
intervertebral_discs.nii.gz
kidney_left.nii.gz
kidney_right.nii.gz
liver.nii.gz
lung_left.nii.gz
lung_right.nii.gz
```

```
pancreas.nii.gz
portal_vein_and_splenic_vein.nii.gz
prostate.nii.gz
sacrum.nii.gz
scapula_left.nii.gz
scapula_right.nii.gz
small_bowel.nii.gz
spinal_cord.nii.gz
spleen.nii.gz
stomach.nii.gz
urinary_bladder.nii.gz
vertebrae.nii.gz
```

Not every written label is meaningful in this field of view. In the full GPU run, 16 of the 50 label masks were non-empty. Of those, 14 are useful candidates for Frac context or landmark work, and 2 should be treated as suspicious/noisy for this crop. The remaining 34 masks were empty out-of-FOV labels written by the generic `total_mr` task.

Tier 1: Direct Artifact Candidate

Mask	Nonzero voxels	Approx. volume	Frac role
`prostate.nii.gz`	50364	37.8 mL	Whole-gland context, registration QA, lesion-search bounds, and future capsular-contact checks

The prostate mask is the only current TotalSegmentator output that should be considered for direct downstream Frac use after visual QA. It is not the FA analysis ROI by itself.

Tier 2: Pelvic Landmarks for Future ROI Proposals

These masks do not give the internal-obturator muscle directly. They are useful as a landmark constellation for a future atlas, registration, or heuristic proposer that targets the internal-obturator calibration sample.

Mask	Nonzero voxels	Approx. volume
`hip_left.nii.gz`	127716	95.8 mL
`hip_right.nii.gz`	126306	94.7 mL
`femur_left.nii.gz`	63171	47.4 mL
`femur_right.nii.gz`	61749	46.3 mL
`urinary_bladder.nii.gz`	125190	93.9 mL
`iliopsoas_left.nii.gz`	53887	40.4 mL

Mask	Nonzero voxels	Approx. volume
`iliopsoas_right.nii.gz`	44688	33.5 mL
`gluteus_maximus_left.nii.gz`	75241	56.4 mL
`gluteus_maximus_right.nii.gz`	46688	35.0 mL
`iliac_artery_left.nii.gz`	3291	2.5 mL
`iliac_artery_right.nii.gz`	3732	2.8 mL
`iliac_vena_left.nii.gz`	4512	3.4 mL
`iliac_vena_right.nii.gz`	4112	3.1 mL

The gluteus-maximus asymmetry is likely influenced by the limited prostate T2 field of view. Treat these as raw evidence and landmarks, not polished user-visible overlays.

Tier 3: Suspicious or Noisy for This Crop

Mask	Nonzero voxels	Approx. volume	Note
`gluteus_medius_left.nii.gz`	477	0.36 mL	Tiny sliver, probably field-of-view edge artifact
`colon.nii.gz`	29136	21.9 mL	Not a reliable bowel/rectum boundary source on this prostate T2 crop

These should not be promoted into Frac outputs. If Frac later needs rectum or bowel context, use a dedicated target or reviewed annotation rather than this generic colon mask.

Empty Out-of-FOV Labels

The other 34 masks are compressed zero masks for labels outside this prostate field of view or not detected here. Examples include upper-body and abdominal labels such as liver, spleen, kidneys, lungs, heart, stomach, pancreas, gallbladder, adrenals, brain, scapulae, clavicles, humeri, vertebrae, spinal cord, and several small muscle labels.

These files are harmless as raw TotalSegmentator output, but they should be filtered before any backend capability registry or user-facing overlay list is generated. Otherwise Frac would overstate what the model actually segmented in this volume.

Missing Targets

Target	TotalSegmentator output?	Frac implication
Internal-obturator muscle ROI	No	Use a DCE-native seeded/landmark proposal, promptable reviewer assistance, or a future small task-specific model on the selected FA reference; do not warp the tiny fixture mask between cases or from T2
Prostate zones, such as PZ/TZ/CG	No	Need a prostate-specific gland/zones model, for example a Prostate158-style nnU-Net
Lesion	No	Need reviewed/imported masks first, then PI-CAI-style detector or lesion segmenter experiments
Tumour-interface ROI	No	Still blocked on the Michallek author rule and a trustworthy lesion contour

Frac Usage Recommendation

For the v1 reviewed-ROI appliance, keep only `prostate.nii.gz` as a candidate direct downstream artifact after visual QA. Its role is registration/context QA, not FA ROI measurement.

Keep the Tier-2 labels in `seg_out/` as raw evidence for future internal-obturator ROI proposal work. Do not promote them to user-visible overlays by default, and do not include them in the normal user-facing `labels` list unless the frontend has a separate landmark-only display path. Do not claim that TotalSegmentator provides the required muscle calibration ROI; at most, it supplies landmarks for a reviewed DCE-space proposal.

For a future backend model-capability entry, TotalSegmentator should be advertised conservatively. The sketch below includes the required field set from fa-backend.md, but it is not a literal schema instance. It also proposes one schema refinement: keep user-visible segmentation labels separate from hidden raw landmark labels so the frontend does not surface landmark masks as normal segmentation claims.

```yaml
id: totalsegmentator-total-mr
kind: segment
engine: TotalSegmentator
status: evaluation
modalities: ["MR"]
anatomy: ["prostate", "pelvis_landmarks"]
labels: ["prostate"]
```

```yaml
license:
  name: Apache-2.0
  commercial_allowed: true
min_vram_gb: 8
supports_4d: false
outputs:
  user_visible_masks: ["prostate"]
  raw_landmark_masks:
    - urinary_bladder
    - hip_left
    - hip_right
    - femur_left
    - femur_right
    - iliopsoas_left
    - iliopsoas_right
    - gluteus_maximus_left
    - gluteus_maximus_right
    - iliac_artery_left
    - iliac_artery_right
    - iliac_vena_left
    - iliac_vena_right
  raw_evidence_files: ["run.json", "gpu-memory.csv"]
version:
  TotalSegmentator: "2.13.0"
  torch: "2.5.1+cu121"
weights:
  source: "~/.totalsegmentator"
  distribution: downloaded-on-first-use
  checksum: TBD
observed_fixture:
  input: "subjects/prostatex-0204/t2.nii.gz"
  gpu: "Quadro T2000"
  peak_vram_mib: 3621
  runtime_s: 62
```

Before this becomes an implementation contract, align the exact field shapes with fa-backend.md: `kind` should use the backend's canonical enum, `labels` may need object entries rather than strings, `outputs` may need to remain a list of output kinds, and `version` / `weights` should follow the backend registry schema. The important policy decision is the split between advertised user-visible masks and raw landmark evidence.

The `min_vram_gb` value above is deliberately more conservative than the 3621 MiB peak observed on this small crop. The measured Martha run proves that this one fixture fits in 4 GB, not that 4 GB is enough for routine pelvic MR segmentation.

That entry should explicitly **not** advertise `muscle_roi`, prostate zones, lesion segmentation, or tumour-interface ROI support. It should also filter empty masks and Tier-3 noisy masks before surfacing anything to the frontend.

GPU Notes

Torch CUDA availability depends on direct GPU device access. In the restricted shell, `torch.cuda.is_availabl` reported `False` with an NVML warning because the sandbox did not expose the NVIDIA device/NVML path fully to Python. With direct device access, the same virtual environment reported:

```
torch:          2.5.1+cu121
torch CUDA:     12.1
CUDA available: True
GPU:            Quadro T2000
```

The full GPU run peaked at 3621 MiB on a 4096 MiB card, approximately 88 percent of reported VRAM. That is acceptable for this small 384 x 384 x 21 T2 fixture, but leaves limited headroom.

Visual QA

The computational run is complete, but the segmentation should not be promoted until it has been visually checked in Frac.

Load the base volume:

```
subjects/prostatex-0204/t2.nii.gz
```

Then load the overlay:

```
seg_out/t2-gpu-full/prostate.nii.gz
```

Acceptance for this memo is modest: confirm that the prostate mask is in the expected anatomical location and follows the gland well enough to be a plausible candidate mask. This does not make TotalSegmentator a validated prostate segmentation method for Frac. It only proves that the local pipeline can produce a mask candidate and provenance for a known T2 fixture.

Next Experiment

The cheapest next experiment is the sequence-independent gland test described in fa-segment.md:

1. run `register-dce` and use its selected calibration-informed FA reference frame
2. run TotalSegmentator `total_mr` on that exact DCE reference frame
3. use `register` to move the T2-derived prostate mask into the canonical FA grid
4. compare overlap and visual alignment

This tests whether TotalSegmentator's whole-gland output is stable enough across MR contrasts to serve as cross-series registration sanity context. It also exercises the required ordering: DCE frames to one reference first, then T2 to that reference. It still does not solve muscle ROI, lesion, or tumour-interface automation.

Reproduction Checklist

1. Create `.venv-seg` and install `requirements-segmentation.txt`.
2. Install `dcm2niix` into `.venv-seg`.
3. Query TCIA for `ProstateX-0204`.
4. Download the `t2_tse_tra` `SeriesNumber 5` UID.
5. Convert DICOM to `subjects/prostatex-0204/t2.nii.gz`.
6. Run `scripts/try_totalsegmentator.sh` on CPU or GPU, using one of the commands in Runs Performed.
7. Inspect `run.json`.
8. Overlay `prostate.nii.gz` on `t2.nii.gz` in Frac.
9. Record visual QA before using the mask in downstream FA work.